

Ethan Yiu

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Data Science student building machine learning models and full-stack applications.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, SQL, JavaScript, TypeScript

Frameworks & Libraries: FastAPI, React, Pandas, NumPy, LangChain

Web & Development Tools: HTML5, CSS3, Tailwind CSS, Git, Linux, Bash, PowerShell

Languages: English, Cantonese, Mandarin, Japanese (native-level written)

EDUCATION

University of Waterloo

Waterloo, ON

*Bachelor of Honours Mathematics, **Data Science** Major*

Sep. 2025 – Present

- GPA: 3.9/4.0
- Relevant Coursework: **Algorithms and Data Abstraction, Linear Algebra, Probability**

PROJECTS

Job Trend Analyzer | *Python, PostgreSQL/SQL, FastAPI, React, Supabase*

May 2026 – Present

- Built a job-market analytics system used by **200+ users**, scraping and cleaning **26,000+ job postings** into structured datasets; developed a FastAPI backend and React frontend.
- Engineered data-quality workflows extracting **70+ in-demand skills**, normalizing salary/role attributes, and powering algorithms that detect emerging skills and salary patterns.
- Optimized backend performance by implementing **asynchronous FastAPI endpoints** and automating daily scraping via **cron-scheduled pipelines**.

Neighbour Intelligence Platform | *FastAPI, Python, Google Maps API, Gemini*

- **Hack Canada 2026:** Built a full-stack livability analysis system across **40+ Waterloo–Kitchener regions**, generating AI-powered reports with cost estimates, commute scoring, and a livability index.
- Implemented async backend **integrating Gemini 2.5 Flash** for natural-language summaries and writing livability scores on-chain via **Solana Web3.py** memo transactions.
- Developed interactive UI featuring a cost heatmap, **radar-chart comparisons**, **address autocomplete**, and responsive dark-mode components built with vanilla HTML/CSS/JS.

Personalized Meal Recommendation | *Auth0, Snowflake, Gemini, ElevenLabs*

- **CXC 2026:** Built a meal-recommendation platform analyzing **audio input**, dietary restrictions, **emotional state**, and past behavior to guide first-year students toward healthier choices.
- Engineered a **Gemini-powered recommendation pipeline**, implemented **Auth0 authentication**, and stored user history in Snowflake; built a responsive **React** interface.

EXPERIENCE

Lead Coding Instructor

May 2026 – Present

Kids Great Minds (KGM)

GTA, ON

- Taught **Python & Java** to **20-30 students** using KGM's curriculum across camps and workshops, covering robotics, **game design**, 3D modeling, and programming tools.
- Debugged student projects by troubleshooting **logic errors**, **software issues**, and hardware setup, ensured over **98% of students** to complete daily coding challenges.
- Coordinated program delivery across **3–5 municipal teaching locations**.

SPORTS ACHIEVEMENTS

Canadian National Table Tennis Athlete (2016–Present); **Top 3 in Canada** (U17 & U19).

Represented BC at 2024 **Olympic Trials**; NCCP Certified Coach.